2^n

11. $f(x) = 1 + 2^x \log 2 \frac{d}{dx} G(x)$

$G(2) = 8$

$G(3) = ?$

Question Incorrect

12. Simple mean question

MLSS 2025 Set 3

Batch ~~2~~ AN

1)

x	0	1	2	3
$P(X)$	0.1	0.2	0.3	0.4

$Var(X) = ?$

$E(X) = \Sigma x \cdot P(X) = 0.2 + 0.6 + 1.2 = 2$

$E(X^2) = \Sigma X^2 P(X) = 0.2 + 1.2 + 3.6 = 5$

$Var(X) = 5 - 2^2 = \boxed{1}$

(11) TO

remember

$\frac{d}{dx}(2^x) = 2^x \ln 2$

$a^x = a^x \ln a$

$\int 2^x \, dx = \frac{2^x}{\ln 2} + c$

MLSS 2025 AN
SET 3

2. 3 fair dice
probability (sum of 3 dice faces is 10)

Total outcomes $= 6^3 = 216$

Favorable outcome:
options:

case 1: (1, 3, 6)
case 2: (1, 4, 5)
case 3: (2, 2, 6)
case 4: (2, 3, 5)
case 5: (2, 4, 4)
Case 6: (3, 3, 4)

case 1: (1, 3, 6)
can be arranged in 3! ways
case 2: (1, 4, 5)
can be arranged in 3! ways
case 4: (2, 3, 5)
can be arranged in 3! ways
case 3: (2, 2, 6)
can be arranged $\frac{3!}{2!}$ ways.
IIIrly case 5: $\frac{3!}{2!}$ ways (one digit repeats twice)

case 6: $3!/2!$ ways
Favorable outcomes $= 3\left(\frac{3!}{2!}\right) + 3 \times 3! = 27$

$\boxed{P = 27/216}$

3. series convergence methods

1) p-series test

$\sum \frac{1}{n^p}$ if $p > 1$ converge
else diverge

2) Limit comparison

Apply limits to $\sum a_n$ and $\sum b_n$

---

5. check book X.

adj(A) when $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\begin{bmatrix} a & -b \\ -c & d \end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

---

6. $f'(x) = 0$ to find critical point
$f''(x)$ substitute critical point
if $f''(x) > 0$, local minimum
if $f''(x) < 0$, local maximum.
if $= 0$, inconclusive

---

8) Fixed Trials with 2 outcomes
→ Binomial Distribution

$$nC_r (p)^r (q)^{n-r}$$

$q = 1-p$

n is 6 tosses / trials,
r is 4 Heads - No. of success

$p = \frac{1}{2}$ (H or T)

Substitute and find answer

Method 2: (By count - not viable by hand)

4 out of 6 Tosses are heads.

$\Rightarrow$ 1) H H H H T T
2) H H H T H T
⋮ and so on

6 C 4 arrangements

Every arrangement has probability

$\left(\frac{1}{2}\right)^6 = \frac{1}{64}$

Ans: $\boxed{\frac{15}{64}}$

11. SD of sampling distribution of Sample Mean = S.E, standard Error

$S.E = \frac{\sigma}{\sqrt{n}}$

12. Popltn Mean 500g
$\mu =$

Sample:
$\bar{x}$ = 492g
$S = 18$
$n = 36$

$$\boxed{\frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}}}$$

Sample mean is 2 below popltn mean
it answer $\boxed{-2}$

13) X be a continuous RV

Interval $[2, 8]$

$P(4 \leq X \leq 6) = ?$

Uniform distribution

Each value in range

EQUALLY LIKELY

$$P(c \leq X \leq d) = \frac{d - c}{b - a}$$

$$= \frac{6 - 4}{8 - 2}$$

$$= \frac{2}{6} = \boxed{\frac{1}{3}}$$

Set 4

MLSS Sample Test

confusion matrix